BankPulse

BANKING TRANSACTION INTELLIGENCE

POWER BI PROJECT DOCUMENTATION

22,253	$1.19M	1,989	$53.64
transactions	transaction amount	unique customers	average amount

A five-page interactive report for exploring transaction activity, customers and cards, merchant performance, time-based patterns, and transaction-level records.

TECHNOLOGY | Microsoft Power BI Desktop | Power Query | DAX | Data Modelling

DATASET | Synthetic transaction data for learning and portfolio demonstration.

01 Project Overview

BankPulse is an interactive Power BI dashboard that turns transaction-level data into a multi-page analytical report. It helps users examine transaction volume and value, customer and card segments, merchant categories, geographic attributes, and time-based activity.

Business objective

Provide one visual workspace for summarising transaction activity, identifying segment differences, monitoring trends, and moving from high-level KPIs to individual transaction records.

Report architecture

Page	Purpose	Key content
Executive Overview	Portfolio-wide summary	Transaction KPIs, monthly trend, transaction amount by income group and age group.
Customer & Card	Customer/card segmentation	Card brands and types, customer demographics, payment channels, top customers/cards and gender split.
Merchant & Geography	Merchant performance and location dimensions	Category and merchant rankings, amount/count comparisons, chip usage, monthly merchant trend.
Time & Intelligence	Time-based patterns	Monthly trend, weekday and hour activity, time-of-day amounts, peak hours, growth and amount bands.
Transaction Data Explorer	Record-level inspection	Year/month/city/state/amount/chip filters and transaction details table.

Portfolio snapshot

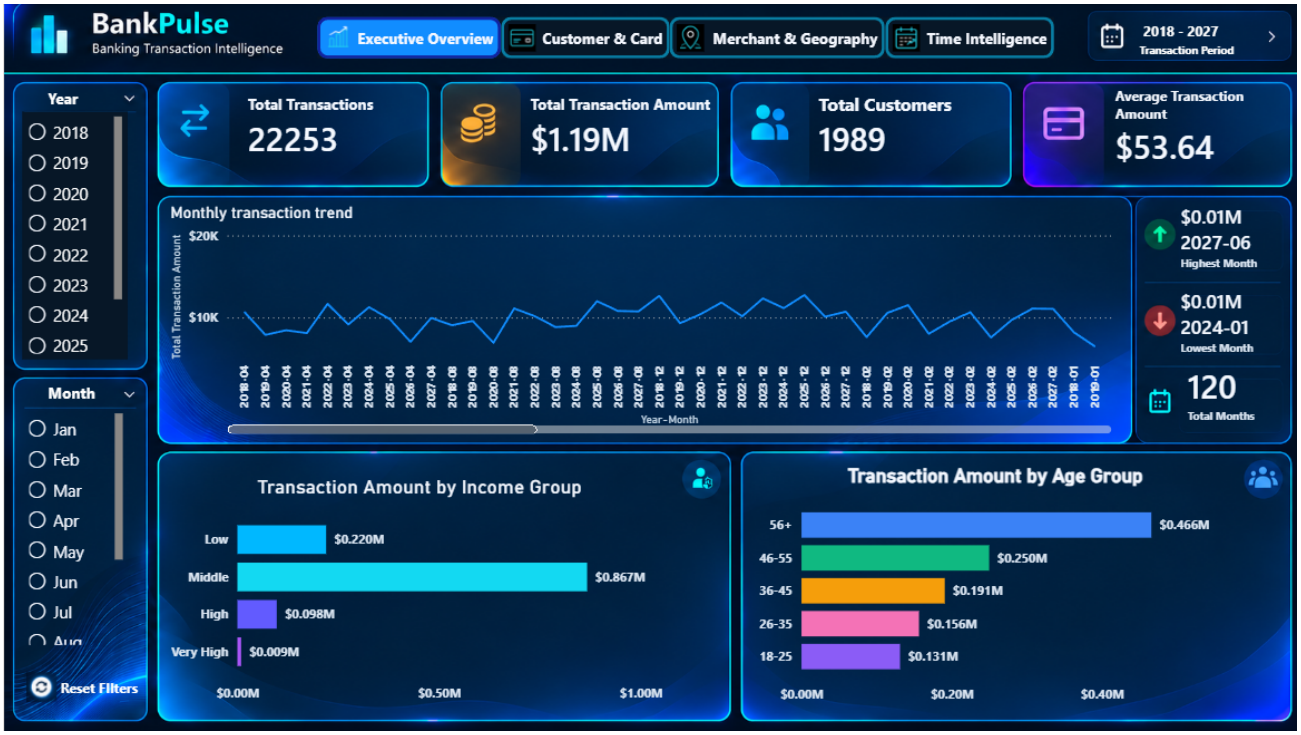
The unfiltered screenshots show 22,253 transactions, approximately $1.19M total transaction amount, an average transaction amount of $53.64, 1,989 customers, 7,341 merchants and 108 merchant categories. These describe the supplied synthetic dataset, not a real bank.

Project purpose

The dashboard demonstrates transaction monitoring, customer segmentation, merchant analysis, time intelligence and detailed record exploration through interactive BI visuals.

02 Executive Overview

The landing page combines headline KPIs with a long-range monthly trend and two segment comparisons.



Supplied screenshot: Executive Overview.

Key metrics

Metric	Meaning
Total Transactions	Count of transaction rows in the current filter context.
Total Transaction Amount	Sum of transaction amounts.
Total Customers	Distinct customer/client identifiers.
Average Transaction Amount	Average transaction amount.

Analytical visuals

The monthly trend compares transaction amounts across the available Year-Month series. The lower charts compare transaction amount by income group and age group. Year and month slicers and a Reset Filters control support focused exploration.

03 Customer & Card

This page examines transaction activity across card brands and types, customer segments and payment channels.



Supplied screenshot: Customer & Card.

Visible KPI cards

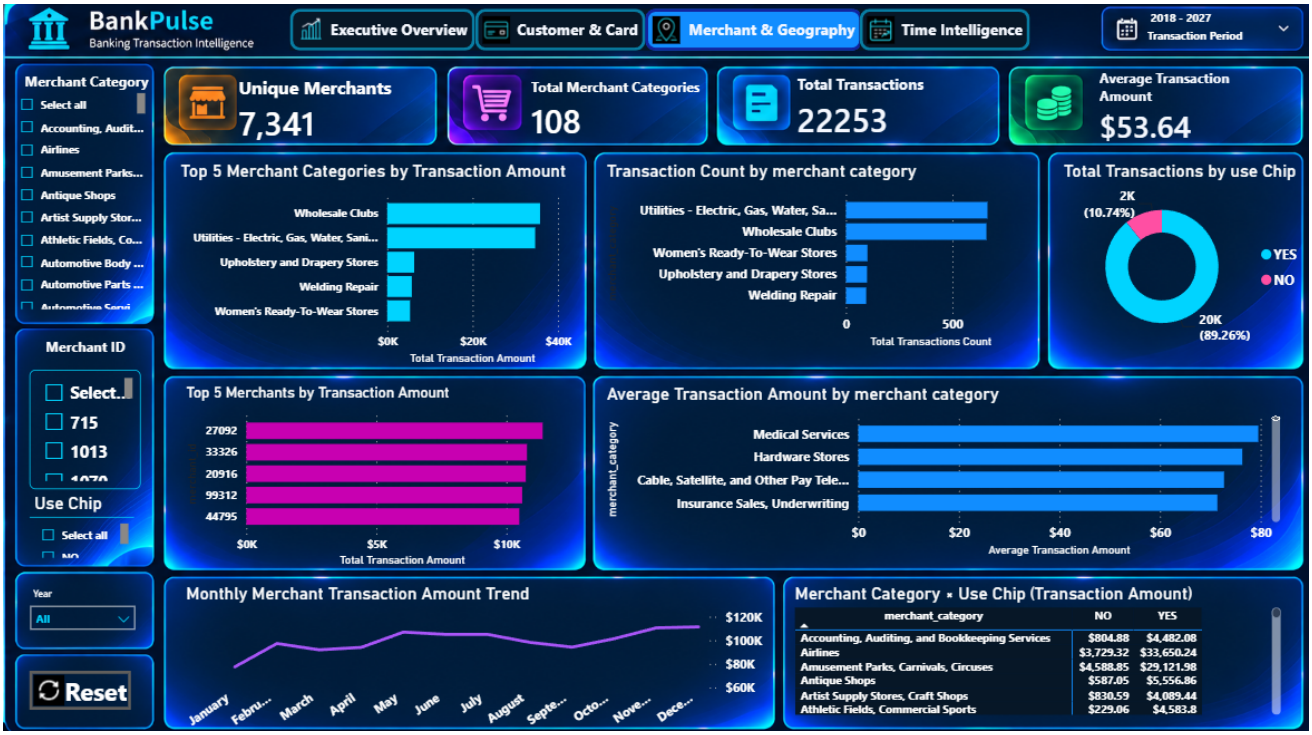
The screenshot reports 4 card brands, 108 merchant categories, 36 state codes represented and 3 card types.

Visuals and use

Card-brand donut and bar charts compare amount and transaction counts. Card-type charts compare transaction amounts. Customer income and age charts describe segment distributions. Additional visuals show transactions by payment channel, top customers and cards by transaction amount, and transaction amount by gender. Slicers include card brand, card type, gender and income group.

04 Merchant & Geography

This page focuses on merchant concentration, category performance and geographic/payment attributes.



Supplied screenshot: Merchant & Geography.

Visible KPI cards

The screenshot shows 7,341 unique merchants, 108 merchant categories, 22,253 transactions and a $53.64 average transaction amount.

Visuals and use

The page compares the top five merchant categories by transaction amount and count, top five merchants by transaction amount, average transaction amount by category, transactions by chip usage, monthly merchant transaction amount, and a merchant-category-by-chip amount matrix. Slicers include merchant category, merchant ID, chip usage and year.

05 Time & Intelligence

This page explores when transactions occur and how transaction volume and amount vary across time.



Supplied screenshot: Time & Intelligence.

Key metrics and visuals

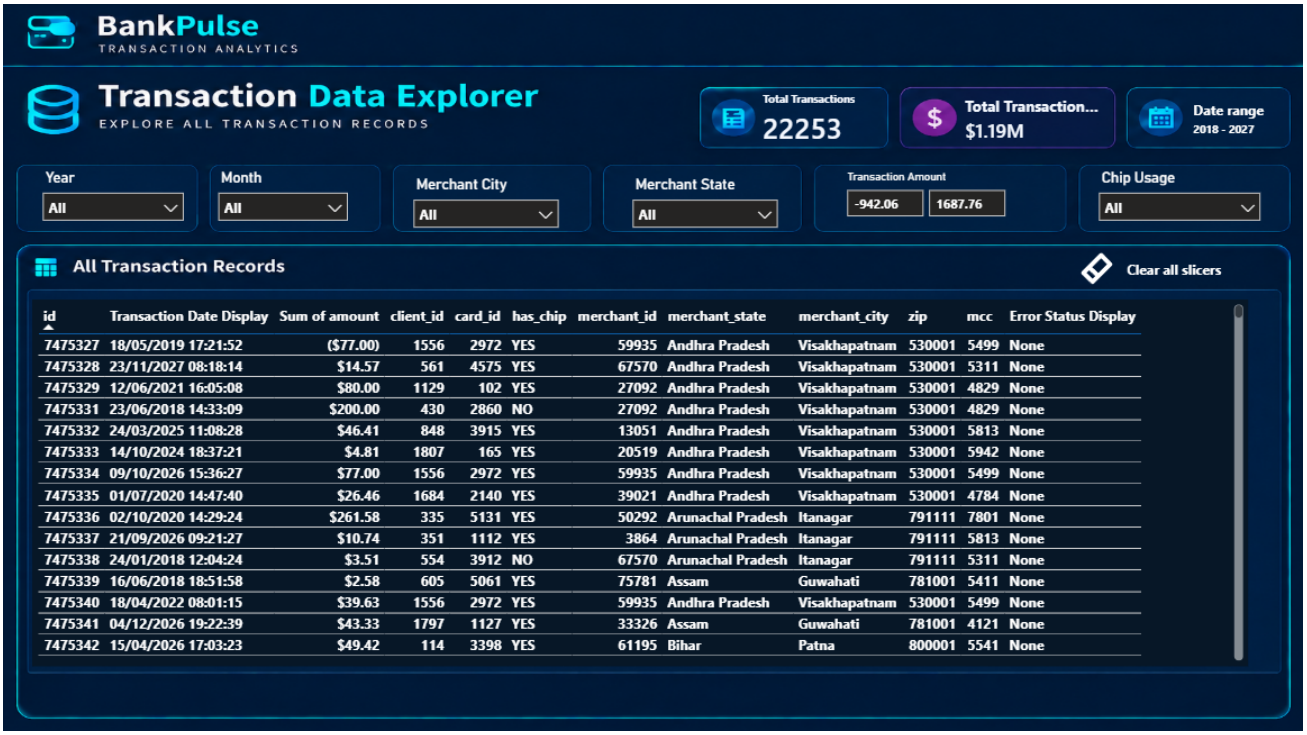
KPIs include transaction count, total amount, average transaction amount, unique customers and unique merchants. Charts cover monthly transactions and amount, weekday counts, hourly counts, amount by time of day, top merchant categories by amount/count, peak transaction hours, monthly growth and transaction amount bands.

Filters

Year, month, weekday, time of day and merchant category slicers allow inspection of different time and category segments. Monthly growth should be read with the Calendar relationship and chronological Year-Month ordering in place.

06 Transaction Data Explorer

The Data Explorer complements aggregated charts with a detailed view of individual transaction records.



id	Transaction Date Display	Sum of amount	client_id	card_id	has_chip	merchant_id	merchant_state	merchant_city	zip	mcc	Error Status Display
7475327	18/05/2019 17:21:52	($77.00)	1556	2972	YES	59935	Andhra Pradesh	Visakhapatnam	530001	5499	None
7475328	23/11/2027 08:18:14	$14.57	561	4575	YES	67570	Andhra Pradesh	Visakhapatnam	530001	5311	None
7475329	12/06/2021 16:05:08	$80.00	1129	102	YES	27092	Andhra Pradesh	Visakhapatnam	530001	4829	None
7475331	23/06/2018 14:33:09	$200.00	430	2860	NO	27092	Andhra Pradesh	Visakhapatnam	530001	4829	None
7475332	24/03/2025 11:08:28	$46.41	848	3915	YES	13051	Andhra Pradesh	Visakhapatnam	530001	5813	None
7475333	14/10/2024 18:37:21	$4.81	1807	165	YES	20519	Andhra Pradesh	Visakhapatnam	530001	5942	None
7475334	09/10/2026 15:36:27	$77.00	1556	2972	YES	59935	Andhra Pradesh	Visakhapatnam	530001	5499	None
7475335	01/07/2020 14:47:40	$26.46	1684	2140	YES	39021	Andhra Pradesh	Visakhapatnam	530001	4784	None
7475336	02/10/2020 14:29:24	$261.58	335	5131	YES	50292	Arunachal Pradesh	Itanagar	791111	7801	None
7475337	21/09/2026 09:21:27	$10.74	351	1112	YES	3864	Arunachal Pradesh	Itanagar	791111	5813	None
7475338	24/01/2018 12:04:24	$3.51	554	3912	NO	67570	Arunachal Pradesh	Itanagar	791111	5311	None
7475339	16/06/2018 18:51:58	$2.58	605	5061	YES	75781	Assam	Guwahati	781001	5411	None
7475340	18/04/2022 08:01:15	$39.63	1556	2972	YES	59935	Andhra Pradesh	Visakhapatnam	530001	5499	None
7475341	04/12/2026 19:22:39	$43.33	1797	1127	YES	33326	Assam	Guwahati	781001	4121	None
7475342	15/04/2026 17:03:23	$49.42	114	3398	YES	61195	Bihar	Patna	800001	5541	None

Supplied screenshot: Transaction Data Explorer.

Filters and table content

Visible filters cover year, month, merchant city, merchant state, transaction amount and chip usage. The table displays a transaction identifier, formatted date/time, amount, client ID, card ID, chip/payment indicator, merchant ID, state, city, ZIP, MCC and simplified error status.

Numeric handling

Use the original numeric amount field for summation and range filtering. Formatted amount text is for display. The supplied screenshot shows row scrolling in the table.

07 Data Model and DAX

Tables in the model

Table	Role
transactions_data	Main transaction fact table: transaction identifiers, timestamp, amount, client/card IDs, merchant IDs, location fields, MCC and error text.
Calendar	Date dimension with date, year, month, quarter, year-month and weekday attributes.
cards_data	Card attributes used in card-level analysis.
users_data	Customer attributes used for age, gender and income-group analysis.
mcc_codes	MCC/merchant-category lookup used for category-level analysis.

Core measures

```
Total Transactions = COUNTROWS('transactions_data')
Total Transaction Amount = SUM('transactions_data'[amount])
Average Transaction Amount = AVERAGE('transactions_data'[amount])
Unique Customers = DISTINCTCOUNT('transactions_data'[client_id])
Unique Merchants = DISTINCTCOUNT('transactions_data'[merchant_id])
```

Monthly growth reference

```
Previous Month Transactions = CALCULATE([Total Transactions], DATEADD('Calendar'[Date], -1, MONTH))
Monthly Growth % = DIVIDE([Total Transactions] - [Previous Month Transactions], [Previous Month Transactions], BLANK())
```

The Calendar-to-transaction relationship should use a date-only transaction field. Retain the original timestamp for hourly analysis. Format Monthly Growth % as Percentage; do not multiply its DAX result by 100.

08 Fields, Design and Interpretation

Key transaction fields

Field	Use in the report
id	Transaction identifier.
date	Transaction timestamp for date/time analysis.
client_id / card_id	Customer and card-level identifiers.
amount	Numeric transaction amount for sums, averages and range filtering.
use_chip	Payment method / chip-use indicator.
merchant_id	Merchant identifier.
merchant_city / merchant_state / zip	Geographic attributes.
mcc	Merchant category code related to the MCC lookup.
errors	Source error text, where available.

Visual design system

Role	Colour	Application
Canvas	#061A33	Dark navy background.
Primary accent	#13C8F5	Cyan borders, active states and highlights.
Secondary accents	#E94BEF / #A020F0	Pink and purple visual differentiation.
Text	#FFFFFF	Titles, labels and values on dark panels.

Interactivity

The report uses consistent navigation across five pages, slicers, KPI cards, chart-level filtering and Reset/Clear Filters controls. Cross-filtering follows the configured model relationships and visual interactions.

Data interpretation

BankPulse uses synthetic practice transaction data. Counts and amounts are dataset-specific and may change with filters. The report demonstrates analytical techniques; it is not evidence of real customer behaviour, fraud rates or banking performance.